APPLICATIONS OF INDOCYANINE GREEN IN VASCULAR SURGERIES: A CASE SERIES

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PURPOSE:

In this study, we report series of TWO cases in vascular surgeries where **Indo Cyanine Green** (ICG) dye had been used.

This provides a direct image visual under fluorescence to confirm the satisfactory blood flow and qualitatively assess the graft/ flap patency and outcomes post-surgery.

The aim of this report is also to discuss various applications and benefits of ICG dye in vascular surgeries.

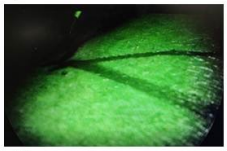
METHOD:

TWO Case reports and Review of Literature.

PRESENTATION OF CASE:

CASE 1: A 68yr male, presented with black discoloration of Left foot 4th and 5th Toes, diagnosed with STENOSIS OF LEFT PROXIMAL AND MID SUPERFICIAL FEMORAL ARTERY. Patient was posted for LEFT FEMORO POPLITEAL BYPASS SURGERY with PTFE graft. Before incision and final closure, 1.5ml (7.5mg) of reconstituted ICG dye was given intravenously (after test dosing) under fluoroscopic vision, images of appearance and peak fluorescence over LEFT foot and LEG were seen on screen and corresponding timings were clocked.

CASE2: A 48yr Male with RIGHT foot and leg Diabetic wet gangrene and crepitus extending till mid thigh, was posted for RIGHT ABOVE KNEE AMPUTATION. Prior to amputation 1.5ml (7.5mg) ICG dye was used to check for the level of satisfactory fluorescence to mark the level of amputation and raising flaps.



RESULT:

CASE1: Peak fluorescence over foot post grafting was noted at a significant **63 s** compared with pre grafting of 92s, indicating satisfactory flow and tissue perfusion immediately after grafting. Post op patient's limb was warm with good volume distal pulsations.

CASE 2: post op D1,3,5 AKA stump was healthy, with no stump infection/ flap necrosis.

DISCUSSION:

Above case were regularly followed up and both the patient had good post op recovery and faster healing and no complications

ICG study enabled immediate visualization of better cutaneous circulation post vascular bypass grafting and also helped in marking the level of amputation where there is better circulation in case of extensive wet gangrene.

Thus intra op ICG use aids in assessing the vascular patency and enable **immediate revision** if necessary

Intra-op ICG study helps in qualitatively assessing better outcome and success of vascular surgeries and avoid early vascular complications post operatively.